

QUARTILES, DECILE, PERCENTILE

Ungrouped data

Quartiles:

When the observation are arranged in increasing order then the values, that divide the whole data into four (4) equal parts, are called quartiles. These values are denoted by **Q1, Q2 and Q3**. It is to be noted that 25% of the data falls below Q1, 50% of the data falls below Q2 and 75% of the data falls below Q3.

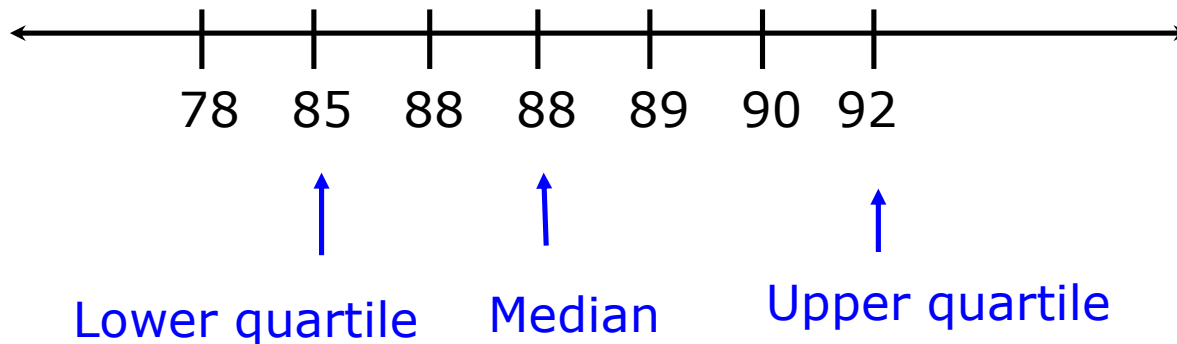
You know that the median of a data set divides the data into a lower half and an upper half. The median of the lower half is the **lower quartile**, and the median of the upper half is the **upper quartile**.

The middle half of the data is represented with a vertical line at the **median**.

The box extends to the **upper and lower quartiles**.

The **upper quartile** is the median of the upper half of the data. The **lower quartile** is the median of the lower half of the data.

Data set: 78, 85, 88, 88, 89, 90, 92



Find the lower and upper quartiles for the data set.

1. Order the data from least to greatest.
2. Find the median.
3. Find the median of the lower half – lower quartile
4. Find the median of the upper half – upper quartile

We do

B. 75, 61, 88, 79, 79, 99, 63, 77

(61 63 75 77) (79 79 88 99) *Order the values.*

lower quartile: $\frac{63 + 75}{2} = 69$

upper quartile: $\frac{79 + 88}{2} = 83.5$

Deciles:

When the observation arranged in increasing order then the values, that divide the whole data into ten (10) equal parts, are called quartiles”

Percentiles:

when the observation are arranged in increasing order then the values, that divide the whole data into hundred(100) equal parts, are called quartiles

Formulas for ungrouped data

Quartile:

Q_j = size of $\left(\frac{j(n+1)}{4}\right)^{th}$ observation, where $j = 1, 2, 3$

Decile:

D_j = size of $\left(\frac{j(n+1)}{10}\right)^{th}$ observation, where $j = 1, 2, \dots, 9$

Percentile:

P_j = size of $\left(\frac{j(n+1)}{100}\right)^{th}$ observation, where $j = 1, 2, \dots, 99$

Question # 1

Calculate Lower Quartile, 5th Decile & 75th Percentile for the following data.

25, 30, 17, 10, 7, 13, 23, 11, 15, 19

Solution

Arrange in ascending order 7, 10, 11, 13, 15, 17, 19, 23, 25, 30

$$Q_1 = \text{first quartile} = \left(\frac{n + 1}{4} \right)^{th} \text{ value}$$

$$Q_1 = \text{first quartile} = \left(\frac{10 + 1}{4} \right)^{th} = 2.75^{th} \text{ value}$$

$$Q_1 = 2^{nd} \text{ value} + 0.75 (3^{rd} \text{ value} - 2^{nd} \text{ value})$$

$$Q_1 = 10 + 0.75 (11 - 10) = 10.75$$

5th Decile

Arrange in ascending order

7, 10, 11, 13, 15, 17, 19, 23, 25, 30

$$D_5 = \left(\frac{5(n+1)}{10} \right)^{th} \text{ value}$$

$$D_5 = \left(\frac{5(10+1)}{10} \right)^{th} = 5.5^{th} \text{ value}$$

$$D_5 = 5^{th} + 0.5 (6^{th} \text{ value} - 5^{th} \text{ value})$$

$$D_5 = 15 + 0.5 (17 - 15) = 16$$

75th Percentile

Arrange in ascending order

7, 10, 11, 13, 15, 17, 19, 23, 25, 30

$$P_{75} = \left(\frac{75(n + 1)}{100} \right)^{th} \text{ value}$$

$$P_{75} = \left(\frac{75(10 + 1)}{100} \right)^{th} = 8.25^{th} \text{ value}$$

$$P_{75} = 8^{th} + 0.25 (9^{th} \text{ value} - 8^{th} \text{ value})$$

$$P_{75} = 23 + 0.25 (25 - 23) = 23.5$$

QUARTILES, DECILE, PERCENTILE

Grouped data

Quartiles

$$Q_j = l + \frac{h}{f} \left(\frac{jn}{4} - c \right), \quad \text{Where } n = \sum f$$

l = the lower boundary of relevant quartile class

n = total frequency

C = the cumulative frequency of the previous interval of relevant quartile class

f = the frequency of relevant quartile class

h = the length of the real interval

Decile

$$D_j = l + \frac{h}{f} \left(\frac{jn}{10} - c \right),$$

$$\text{Where } n = \sum f$$

Percentile

$$P_j = l + \frac{h}{f} \left(\frac{jn}{100} - c \right),$$

$$\text{Where } n = \sum f$$

Continuous series (grouped data)

Step1: Find cumulative frequencies

Step2: Find $\frac{n}{4}$

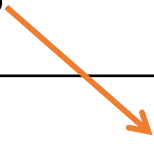
Step3: See in the cumulative frequencies, the value just greater than $\frac{n}{4}$, then the Corresponding class interval is called **first quartile class**. Then apply the respective formulae.

For **2nd Quartile** find $\frac{2n}{4}$, See in the cumulative frequencies, the value just greater than $\frac{2n}{4}$, then the Corresponding class interval

For **3rd Quartile** find $\frac{3n}{4}$, See in the cumulative frequencies, the value just greater than $\frac{3n}{4}$, then the Corresponding class interval

Example: find the quartiles Q1, Q2, and Q3 of the following data

Class	Frequency
30-40	8
40-50	87
50-60	190
60-70	304
70-80	211
80-90	85
90-100	20

Class	Frequency	Cumulative frequency
30-40	8	8
40-50	87	95 
  50-60	190 	285
60-70	304	589
70-80	211	800
80-90	85	825
90-100	20	905 

Calculate $\frac{n}{4} = \frac{905}{4} = 226.25$ (for 1st class interval)

$$\sum_{i=1}^n f_i = 905$$

We know that $Q_j = l + \frac{h}{f} \left(\frac{jn}{4} - c \right),$

Put j=1

$$Q_1 = l + \frac{h}{f} \left(\frac{n}{4} - c \right)$$

l = the lower boundary of relevant quartile class =50

n = total frequency= 905

C = the cumulative frequency of the previous interval of relevant quartile class =95

f = the frequency of relevant quartile class =190

h = the length of the real interval =10

$$Q_1 = 50 + \frac{10}{190} \left(\frac{905}{4} - 95 \right) = 56.82$$

For 2nd Quartile Put j=2

$$Q_2 = l + \frac{h}{f} \left(\frac{2n}{4} - c \right)$$

$$\text{find } \frac{2n}{4} = \frac{2 \times 905}{4} = 452.5$$

Check in cumulative frequency column

Class	Frequency	Cumulative frequency
30-40	8	8
40-50	87	95
50-60	190	285
 ← 60-70	304 → 	589
70-80	211	800
80-90	85	825
90-100	20	905

l = the lower boundary of relevant quartile class =60

n = total frequency= 905

C = the cumulative frequency of the previous interval of relevant quartile class =285

f = the frequency of relevant quartile class =304

h = the length of the real interval =10

$$Q_1 = l + \frac{h}{f} \left(\frac{2n}{4} - c \right)$$

$$Q_2 = 60 + \frac{10}{304} (452.5 - 285) = 65.36$$

Calculate the 4th and 8th Decile

$$Q_j = l + \frac{h}{f} \left(\frac{jn}{10} - c \right), \quad \text{Put } j = 4$$

$$Q_j = l + \frac{h}{f} \left(\frac{4n}{10} - c \right)$$

For 4th Decile find $\frac{4n}{10} = \frac{4 \times 905}{10} = 362$

The interval where the 4th decile lies has the cumulative frequency of 589.

Class	Frequency	Cumulative frequency
30-40	8	8
40-50	87	95
50-60	190	285
  60-70	304  f	589
70-80	211	800
80-90	85	825
90-100	20	905

$$D_5 = 60 + \frac{10}{304} (452.5 - 285) = 65.36$$

Calculate the 18th and 76th percentile

$$P_j = l + \frac{h}{f} \left(\frac{jn}{100} - c \right)$$

For 18th percentile find $\frac{18n}{100} = \frac{18 \times 905}{100} = 162.9$

The interval where the 18th percentile lies has the cumulative frequency of 285

Class	Frequency	Cumulative frequency
30-40	8	8
40-50	87	95 → 
 ← 50-60	190 → 	285
60-70	304	589
70-80	211	800
80-90	85	825
90-100	20	905

$$\mathbf{P}_{18} = 50 + \frac{10}{190}(162.9 - 95) = 125.43$$

Quartile Deviation or Semi-inter-quartile Range:

The half of the difference between the upper quartile and lower quartile is called the semi-inter quartile range or quartile deviation "i.e.

$$\text{Quartile deviation} = \frac{Q_3 - Q_1}{2}$$

Coefficient of Quartile Deviation:

The coefficient of quartile deviation is a relative measure of dispersion and is given by:

$$\text{Coefficient } Q.D = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$